MarketSight

Stock Market Prediction System

Project Plan

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COSC 471 - Winter 2026 | Okanagan College

March 17, 2026

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1. Introduction

1.1 Project Overview

MarketSight is a web-based stock market prediction platform developed for COSC 471 at Okanagan College. The system leverages machine learning (XGBoost with Optuna HPO) to predict stock prices using 53 technical indicators, presenting results through a React dashboard and Streamlit data explorer.

1.2 Project Objectives

- Deliver an end-to-end ML prediction pipeline from data ingestion to user interface
- Provide prediction confidence scores to help users assess forecast reliability
- Deploy to production infrastructure (DRI server) with automated pipelines
- Achieve Initial Operational Capability (IOC) by end of Sprint 5

1.3 Methodology

The project follows the **Agile Unified Process (AUP)** with Scrum ceremonies. Development is organized into 1-3 week sprints with XP pair programming. The team conducts sprint planning, daily standups, backlog grooming, sprint reviews, and retrospectives.

2. Project Organisation

2.1 Team Structure

Role	Member	Responsibilities
Scrum Master	Zane Tessmer	Sprint ceremonies, documentation, coordination, impediment removal
Product Owner	Harsh Kumar	Backlog prioritization, architecture decisions, stakeholder communication
Lead Developer	Harsh Kumar	Backend, ML pipeline, infrastructure, DevOps
Developer	Dante Bertolutti	Confidence scoring, predictions UI, acceptance testing

Developer	Parag Jindal	Frontend routing, API client, type safety
Developer	Kaval S	User settings, dashboard optimization
Developer	Guntash	Dashboard cards, UI polish, accessibility

2.2 XP Pairs by Sprint

Sprint 3 (Jan 20 - Feb 9, 2026)

Pair	Members	Focus
Pair 1	Harsh & Parag	Backend API + Frontend scaffold
Pair 2	Dante & Kaval	ML notebook + Database schema
Pair 3	Guntash & Zane	CI/CD + Documentation

Sprint 4 (Feb 10 - Mar 3, 2026)

Pair	Members	Focus
Pair 1	Harsh & Dante	Inference API + Confidence scoring
Pair 2	Parag & Guntash	Frontend routing + Dashboard cards
Pair 3	Kaval & Zane	User settings + Documentation

Sprint 5 (Mar 4 - Mar 17, 2026)

Pair	Members	Focus
Pair 1	Harsh & Dante	ML pipeline integration + Confidence rebase
Pair 2	Parag & Kaval	Frontend cleanup + Dashboard optimization
Pair 3	Guntash & Zane	UI polish + Documentation + Testing

3. Risk Analysis and Mitigation

ID	Risk	Probability	Impact	Severity	Mitigation Strategy	Owner
R-01	Model accuracy insufficient for useful predictions	Medium	Medium	Medium	53-feature pipeline, Optuna HPO, continuous tuning	Harsh
R-02	No authentication exposes endpoints	Medium	High	High	CORS restriction now; JWT auth in backlog	Harsh
R-03	DRI/DRAC server downtime	Low	High	Medium	Management scripts, monitoring, local fallback	Harsh
R-04	FMP API rate limiting or outage	Low	Medium	Low	Chunked downloads, gap-filling scripts, cached data	Harsh
R-05	Merge conflicts between major feature branches	Medium	Medium	Medium	Merge coordination meetings, branch protection	Zane
R-06	Team member availability fluctuations	Medium	Medium	Medium	Cross-training via XP pairs, documented code	Zane
R-07	Scope creep from stakeholder requests	Low	Medium	Low	Product backlog prioritization, sprint boundaries	Harsh
R-08	Database growth exceeding storage	Low	Low	Low	Data retention policies, indexing, monitoring	Harsh
R-09	Frontend/backend API contract mismatches	Medium	Low	Low	TypeScript types matching Pydantic schemas	Parag
R-10	CI pipeline failures blocking merges	Low	Medium	Low	Fix-forward policy, parallel job execution	Harsh

Risk Severity Matrix

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              Low Impact    Medium Impact    High Impact
High Prob     |             |                |
Medium Prob   | R-09        | R-01, R-05,    | R-02
              |             | R-06, R-07     |
Low Prob      | R-04, R-08  | R-03, R-08,    |
              |             | R-10           |
```

4. Work Breakdown Structure

4.1 WBS Hierarchy

```
MarketSight
|
+-- 1. Backend
|   +-- 1.1 API Layer (routers, health check, dependency injection)
|   +-- 1.2 Market Module (stock models, CRUD, service, endpoints)
|   +-- 1.3 Inference Module (model loading, features, confidence, prediction API)
|   +-- 1.4 Training Module (training jobs API, log streaming)
|   +-- 1.5 Data Module (snapshot API: build, list, download)
|   +-- 1.6 Core (config, database, security)
|
+-- 2. Frontend (React)
|   +-- 2.1 Dashboard (summary cards, layout)
|   +-- 2.2 Stocks (list, search, detail, charts)
|   +-- 2.3 Predictions (selector, predict, batch, cards)
|   +-- 2.4 Training Monitor (log viewer, progress)
|   +-- 2.5 Admin (user management)
|   +-- 2.6 Settings (profile, password, account)
|   +-- 2.7 Services & Hooks (API clients, state management)
|
+-- 3. Frontend (Streamlit)
|   +-- 3.1 Market Data Explorer
|   +-- 3.2 Inference Testing
|   +-- 3.3 Dataset Management
|
+-- 4. ML Pipeline
|   +-- 4.1 Data Ingestion (FMP API, gap filling)
|   +-- 4.2 Feature Engineering (53 indicators)
|   +-- 4.3 Model Training (XGBoost, Optuna HPO)
|   +-- 4.4 Model Artifacts (native .ubj, meta.json)
|   +-- 4.5 DRAC Pipeline (end-to-end training)
|
+-- 5. Infrastructure
|   +-- 5.1 Docker Compose (all services)
|   +-- 5.2 Nginx (reverse proxy, SSL)
|   +-- 5.3 Airflow (DAGs: seed, retrain, snapshot)
|   +-- 5.4 CI/CD (GitHub Actions)
```

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|   +-- 5.5 DRI Server (deployment, management scripts)
|
+-- 6. Testing
|   +-- 6.1 Backend Unit Tests
|   +-- 6.2 Backend Integration Tests
|   +-- 6.3 Acceptance Tests
|   +-- 6.4 CI Pipeline Validation
|
+-- 7. Documentation
    +-- 7.1 Sprint Ceremonies & Backlogs
    +-- 7.2 Technical Documentation
    +-- 7.3 User Support (Manual, Help, Installation)
    +-- 7.4 Developer's Guide
```

5. Project Schedule and Resource Allocation

5.1 Sprint Schedule

Sprint	Phase	Duration	Dates	Velocity
Sprint 3	Construction 1	3 weeks	Jan 20 - Feb 9, 2026	30 pts
Sprint 4	Construction 2	3 weeks	Feb 10 - Mar 3, 2026	73 pts
Sprint 5	Construction 3	2 weeks	Mar 4 - Mar 17, 2026	62 pts

5.2 Resource Allocation (Sprint 5 - Current)

Team Member	Hours/Week	Sprint 5 Total	Stories	Points
Harsh Kumar	25 hrs	50 hrs	US-32 to US-38	36
Dante Bertolutti	15 hrs	30 hrs	US-30, US-31, US-40	15
Zane Tessmer	12 hrs	25 hrs	US-39, US-40	8
Parag Jindal	10 hrs	20 hrs	US-41	3
Kaval S	7 hrs	15 hrs	US-42	3
Guntash	7 hrs	15 hrs	US-43	2
Total	**76 hrs**	**155 hrs**	**14 stories**	**67 pts**

5.3 Cumulative Story Points

Sprint	Planned	Completed	Cumulative
Sprint 3	30	30	30
Sprint 4	73	73	103
Sprint 5	62	62	165

6. Project Milestones

Milestone	Date	Description	Status
M1: Project Kickoff	Jan 13, 2026	Team formed, roles assigned, repo created	Achieved
M2: Architecture Baseline	Jan 20, 2026	Tech stack selected, Docker env running	Achieved
M3: Sprint 3 Complete	Feb 9, 2026	Core infrastructure, ML notebook, initial frontend	Achieved
M4: Sprint 4 Complete	Mar 3, 2026	Inference API, confidence scoring, Airflow, FMP pipeline	Achieved
M5: Sprint 5 Complete (IOC)	Mar 17, 2026	Production deployment, acceptance tests, client sign-off	Achieved
M6: Final Submission	Mar 18, 2026	All deliverables submitted to Moodle	Pending

7. Gantt Chart

```
Task                          Jan    Feb             Mar
                              W3 W4  W1 W2 W3 W4    W1 W2 W3
Sprint 3                      |======|======|======|
  Backend scaffold            |======|
  ML training notebook              |======|
  Frontend scaffold                  |======|
  CI/CD pipeline              |======|
  Docker environment          |======|

Sprint 4                                  |======|======|======|
  Inference API                           |======|
  Market module                           |======|
  FMP data integration                    |======|======|
  53-feature pipeline                            |======|
  Confidence scoring                             |======|
  Predictions page                                      |======|
  Airflow DAGs                            |======|
  Optuna HPO + DRAC                              |======|======|
  Backend tests                           |======|

Sprint 5                                                     |======|======|
  Confidence rebase                                          |======|
  Predictions restore                                               |======|
  Streamlit frontend                                         |======|
```

```
DRI server migration                                          |======|======|
Model format migration                                        |======|
Feature eng refactor                                          |======|======|
DW sync scripts                                                      |======|
Documentation                                                 |======|======|
Acceptance testing                                                   |======|
```

8. Monitoring and Reporting

8.1 Scrum Ceremonies

Ceremony	Frequency	Duration	Facilitator
Daily Standup	Daily (class days)	15 min	Zane Tessmer
Sprint Planning	Start of sprint	20 min	Zane Tessmer
Backlog Grooming	Mid-sprint	20 min	Zane Tessmer
Sprint Review	End of sprint	15 min	Zane Tessmer
Sprint Retrospective	End of sprint	15 min	Zane Tessmer

8.2 Reporting Mechanisms

Report	Frequency	Audience	Format
Daily Scrum report	Per standup	Moodle submission	PDF (pair-based)
Sprint backlog burndown	Weekly	Team	Screenshot
Sprint review notes	End of sprint	Client + Professor	PDF
Performance schedule	Per sprint	Assistant Instructor	Email
Individual report	End of sprint	Moodle	PDF

8.3 Issue Tracking

- **Platform:** GitHub Issues (labels: `ml-pipeline`, `frontend`, `testing`, `api`)
- **PR Reviews:** Required before merge to `main`
- **CI Dashboard:** GitHub Actions (6 jobs: lint, typecheck, test, lint, build, docker)

8.4 Communication Channels

Channel	Purpose
GitHub Issues	Bug tracking, feature requests

GitHub Pull Requests	Code review, merge coordination
In-class meetings	Standup, sprint ceremonies
Direct messages	Urgent blockers, pair coordination